K. Çağrı Kuzaytepe

Software Engineer | ML Developer | Full-Stack Developer

kuzaytepecagri@gmail.com | github/cagrikuzaytepe | linkedin/cagrikuzaytepe | Ankara, Turkey

Professional Summary

Computer Engineering student at Hacettepe University with extensive hands-on experience building production-quality software across three disciplines: **full-stack web engineering** (React + .NET / Spring Boot), **deep learning & NLP** (PyTorch, HuggingFace, LangGraph, RAG pipelines), and **distributed systems** (event-driven microservices, Docker, Clean Architecture). Delivered real-time collaborative tools with SignalR, a fully offline RAG agent with near-zero hallucination, and a 4-module computer vision research series covering homography, colorization, and transformer-based object detection. Currently serving as a Python instructor and technical mentor at the IEEE Hacettepe Student Branch.

Technical Skills

Languages	Python · C# · Java · TypeScript · JavaScript · Dart · SQL
Backend	ASP.NET Core 8/9 · Spring Boot · FastAPI · Entity Framework Core · Clean Architecture · CQRS / MediatR
Frontend	React.js 18/19 · TypeScript · Tailwind CSS · Vite · Zustand · Flutter · Gradio · Streamlit
ML / AI	PyTorch · HuggingFace Transformers · LangGraph · OpenCV · scikit-learn · FastText · NLTK · Sentence-Transformers
Databases	PostgreSQL · SQL Server · Redis · Vespa (Vector DB) · MSSQL
DevOps	Docker · Docker Compose · RabbitMQ · SignalR · GitLab CI/CD · Prometheus · Swagger / OpenAPI

Featured Projects

Full-Stack Web Engineering

DeskFlow — Intelligent Office Desk Booking System *2024 – 2025*

.NET 8 | React 18 | TypeScript | SignalR | Konva.js | SQL Server | Docker | Clean Architecture

- Designed and built an **interactive floor plan editor** using Konva.js canvas — admins drag-and-drop desks to construct or update real office layouts without any hardcoded maps.
- Implemented **three fully independent booking modes**: single-day reservation, recurring weekly slots (e.g., every Monday and Wednesday), and team bulk-booking where a manager reserves a cluster of desks for their entire team in one action.
- Integrated **SignalR WebSockets** so every desk status change propagates instantly across all connected browser sessions — no polling, no page refresh required.
- Built a secure **QR code check-in** system: each physical desk has a unique QR code; scanning activates the reservation. A .NET hosted background service monitors all active bookings and **auto-cancels no-shows within 15 minutes**, freeing desks automatically.
- Structured the entire backend on **Clean Architecture** principles — Domain layer carries zero external dependencies; Application layer defines CQRS-style use cases; Infrastructure handles EF Core and SQL Server; API layer exposes controllers and SignalR hubs.
- Enforced **role-based access control** with JWT bearer tokens: employees see only their own bookings and the visual floor plan, while admins access a full analytics dashboard (occupancy trends, most-booked desks, user activity charts via Recharts).

DevBoard — Real-Time Kanban & Project Management App *2024 – 2025*

.NET 9 | React 19 | TypeScript | PostgreSQL | Redis | SignalR | CQRS / MediatR | dnd-kit

- Developed a spatial, window-based productivity tool with a three-level hierarchy — **Workspace → Project → Issue** — each navigable from a floating panel UI that mimics a desktop operating environment.
- Applied **CQRS + MediatR**: every write operation (CreateIssue, MoveCard, UpdateStatus) goes through a dedicated command handler; reads go through optimized query handlers — zero bleed-through between the two paths.
- Reduced PostgreSQL pressure on high-frequency reads (e.g., full board state on load) with **Redis distributed caching**; cache invalidation is triggered automatically on write commands via MediatR pipeline behaviors.
- Synchronized all connected clients through **SignalR hubs**: when a user drags a card between columns, every other

session watching the same board sees the change in real time.
- Secured all endpoints with **JWT authentication and BCrypt password hashing**; implemented refresh token rotation to maintain session continuity without re-login.
- Built the drag-and-drop board using **dnd-kit**, the floating panel windows and overlapping UI using custom Zustand state slices.

Pack & Eat — Multi-Role Food Delivery Platform *2024*
ASP.NET Core 8 | React.js | Tailwind CSS | SQL Server | EF Core | Docker | Supabase | Google Maps API

- Delivered a complete, production-ready food delivery ecosystem with **four distinct role panels** in a single monorepo: customer ordering flow, restaurant order management, courier delivery tracking, and admin oversight.
- Embedded **live courier location tracking** using the Google Maps JavaScript API — customers see a moving marker showing the courier's exact GPS position as the order travels.
- Integrated **Supabase Storage** for all restaurant images and menu photos, using pre-signed URLs for secure, direct browser-to-storage access without routing media through the API server.
- Implemented **JWT-based multi-role authentication**: the same login endpoint issues role-scoped tokens that redirect users to their corresponding dashboard automatically.
- Set up a **GitLab CI/CD pipeline** — automated build, test, and Docker image build jobs run on every push; the Postman collection was maintained as a regression test suite run in the pipeline.

Backend, Microservices & Systems Architecture

Mini E-Commerce — Event-Driven Microservices Platform *2024*
Spring Boot | Java | PostgreSQL | RabbitMQ | Docker Compose | Prometheus | Swagger

- Architected **six fully decoupled microservices** — Order (8001), Payment (8002), Stock (8003), Cargo (8004), Notification (8005), and API Gateway (8080) — each with its own Spring Boot application, database schema, and lifecycle.
- Implemented the **choreographed saga pattern**: no central orchestrator exists; services react to domain events published on RabbitMQ queues, creating a resilient event chain (order placed → payment processed → stock deducted → tracking code generated → customer notified).
- Designed **compensating transaction logic**: if stock deduction fails after payment succeeds, a reversal event triggers the Payment service to refund automatically.
- Exposed **Prometheus metrics** via Spring Boot Actuator on each service; the RabbitMQ Management UI was used to monitor queue depths and consumer lag in development.
- Documented every service with **Swagger/OpenAPI**; the API Gateway aggregates docs from all downstream services into a single browsable UI.

UAV Control Application — Drone Automation System *2024*
Python | FastAPI | Uvicorn | Drone Telemetry | RESTful API

- Engineered a comprehensive drone control system with a **dual-interface approach**: a desktop GUI for manual, real-time piloting and monitoring, and a REST API for programmatic control and integration with third-party systems.
- Implemented endpoints for all critical flight operations: `POST /connect`, `/arm`, `/takeoff`, `/land`, `/goto` (GPS coordinates), `/set_mode`, and `GET /get_telemetry` — all using standardized JSON request/response bodies.
- Streamed real-time telemetry (altitude, airspeed, GPS position, heading, battery voltage and level) to connected clients; telemetry polling runs as a background thread to avoid blocking the API event loop.
- Separated the codebase into three independent modules — API layer, GUI controller, and core drone communication logic — following **separation of concerns** to allow hardware swapping without modifying the interface layer.

University Social Media Backend — Student Networking API *2024*
.NET 8 | ASP.NET Core Web API | Entity Framework Core | SQL Server | JWT

- Built the core backend infrastructure for a university-focused social networking platform using an **N-Tier architecture** (Controllers, Services, Repositories, Data).
- Modeled relational entities — Users, Posts, Comments, Follows, Reactions — and managed all schema evolution through **EF Core code-first migrations**.
- Secured all endpoints with **JWT bearer authentication**; documented the full API surface with Swagger for front-end team consumption.

Machine Learning, NLP & Data Science

Self-RAG Agent — Fully Offline Secure Document Analysis *2024*
LangGraph | Llama (4-bit Quantized) | Vespa | Sentence-Transformers | Streamlit

- Architected a **fully local, offline RAG system** — zero external API calls; all inference, retrieval, and generation run on-device, making it suitable for air-gapped or privacy-critical environments.
- Engineered an **agentic LangGraph workflow** with three automated quality gates: (1) document relevance grading — irrelevant retrieved chunks are discarded before generation; (2) adaptive query rewriting — if the retrieved context is insufficient, the agent reformulates the question and retries; (3) final self-critique — the generated answer is validated against the source before delivery. This loop **reduces hallucination to near zero**.

- Implemented **semantic chunking** — documents are split at semantic boundaries rather than fixed token windows, preserving meaningful context units and improving retrieval precision significantly.
- Indexed all document vectors into a locally deployed **Vespa** instance using Sentence-Transformers embeddings; achieved **millisecond-level semantic search** entirely on-premise.
- Integrated a **4-bit quantized Llama** model configured for thread-safe concurrent query handling; the quantization reduces memory footprint while preserving generation quality for document Q&A tasks.
- Built a **Streamlit control panel** covering the full operational lifecycle: document upload and chunking, indexing status monitoring, query history, retrieval debug views, and system performance metrics.

Image Classification — CNNs & Transfer Learning Comparison *2024*

PyTorch | ResNet18 | MobileNetV2 | Data Augmentation | Animal-10 Dataset

- Designed a **custom CNN from scratch** — four convolutional blocks with batch normalization, ReLU activations, max pooling, and adaptive average pooling into a fully connected classifier.
- Fine-tuned **pretrained ResNet18** (74% accuracy) and **MobileNetV2** (83% accuracy, best overall) by replacing their classification heads and unfreezing the final layers for domain adaptation on the 10-class Animal dataset.
- Applied an aggressive **data augmentation pipeline** — random rotations, horizontal flips, color jitter, and normalization — to combat overfitting on the limited dataset.
- Produced a structured **comparative analysis**: training curves, per-class confusion matrices, and precision/recall/F1 breakdowns for all three model families; documented where transfer learning outperforms training from scratch and why.

LSTM Sentiment Analysis, NLP Pipelines & Recommendation Systems *2023 – 2024*

PyTorch | FastText | HuggingFace Transformers | NLTK | scikit-learn | Gradio | FastAPI

- **LSTM IMDb Classifier:** Trained a custom LSTM (dropout, batch normalization, FC output) initialized with **pretrained FastText 300-dim embeddings** on 10,000 balanced IMDb reviews, achieving **87%+ F1-score**. Explicitly handled OOV words and ran a head-to-head comparison against PyTorch's standard LSTM across accuracy, precision, recall, and F1.
- **Book Recommendation System:** Built a multi-stage NLP pipeline — sentiment classification of book descriptions (to extract emotional tone), vector embedding of descriptions using HuggingFace models, and a **semantic vector search** layer to rank books by similarity to the user's free-text query. Deployed as an interactive **Gradio dashboard**.
- **Movie Sentiment Recommender:** Used **NLTK VADER** for real-time sentiment scoring of user mood input; applied **scikit-learn content-based filtering** to map detected sentiment to matching movies. Served via a **FastAPI** backend with a plain HTML/JS frontend, fully containerized with **Docker Compose**.
- **Text Summarization:** Fine-tuned and evaluated **HuggingFace sequence-to-sequence transformer models** on a custom dataset; integrated a full data preprocessing and training pipeline with ROUGE-metric evaluation.

Advanced Computer Vision

Computer Vision Project Series — 4 Research Modules *2023 – 2024*

PyTorch | OpenCV | U-Net | DETR | RANSAC | SIFT / SURF | PatchGAN | VGG16

- **Generalized Hough Transform:** Implemented Canny edge detection, R-Table construction, a 4-dimensional voting accumulator (position x, y + scale + rotation), and Non-Maximal Suppression from scratch. The system detects arbitrary template shapes in cluttered scenes robust to scale and rotation changes.
- **Homography, Panorama Stitching & Augmented Reality:** Extracted SIFT keypoints and descriptors from multiple images, matched them with a ratio test, then computed a robust homography matrix using the **DLT algorithm and RANSAC** (implemented from scratch) to reject outlier matches. Used the homography to stitch seamless panoramas and, in a second task, to warp and project a live AR video clip onto a physical book cover detected in a playing video, with alpha blending for smooth compositing.
- **Image Colorization with Deep Learning:** Converted grayscale images to the **CIELAB color space**, passing the L* channel through four progressively advanced architectures: (1) a plain encoder-decoder CNN, (2) a **U-Net with skip connections** preserving spatial detail, (3) a **pretrained ResNet18 backbone** for richer feature extraction, and (4) a **PatchGAN adversarial setup** for perceptually realistic colors. Also experimented with **VGG16 perceptual loss** to penalize structural dissimilarity.
- **DETR Object Detection:** Implemented a simplified DEtection TRansformer — a CNN backbone feeding a standard transformer encoder-decoder — trained on the PennFudan pedestrian dataset. Implemented **bipartite matching (Hungarian Algorithm)** for one-to-one prediction-to-ground-truth assignment and **set-based loss** combining GIoU bounding box loss with classification cross-entropy. Conducted a full **ablation study**: compared ResNet18 vs. MobileNet backbones and varied the number of object queries, measuring impact on mAP.

Mobile Development & Security

ARK4UTH Authenticator — Secure Cross-Platform Mobile App *2024*

Flutter | .NET | MSSQL | Redis | AES-256 Encryption | Push Notifications

- Developed a cross-platform (iOS and Android) mobile authentication application with a responsive **Flutter** frontend and a multi-layered **.NET** backend designed for scalability.
- Secured all user credentials and data in transit using **end-to-end AES-256 encryption**; built a key management infrastructure to handle encryption key rotation.

- Optimized session and token lookup latency with **Redis caching**, keeping hot authentication data in memory to avoid repetitive database hits.
- Implemented a **real-time push notification system** that fires an alert to the user's device on every authentication attempt, enabling instant approval or denial without manual polling.
- Maintained **comprehensive structured logging** of every authentication event for auditability and anomaly detection, with alerting logic for brute-force or unusual access patterns.

Education Technology

Personal Portfolio Website v2 *2025*
React.js | CSS Modules | Component-Based Architecture

- Built a fully component-driven personal portfolio website using **React.js** with modular CSS; sections include Hero, About, Experience, Skills, Projects, Events, and Contact.
- Applied responsive design principles for seamless rendering across mobile, tablet, and desktop viewports.

Volunteer

Python Instructor & Technical Mentor *2025 – 2026*
IEEE Hacettepe Student Branch — ProgrammixV Training Series Ankara, Turkey

- Sole designer and instructor of a **3-week project-oriented Python curriculum** delivered to 30+ university students: covered development environment setup, variables and data types, operators, I/O, conditional logic, loops (for/while) with break/continue, and file/data management (reading and writing .txt and .json files).
- Prepared all course materials independently — weekly lecture presentations, timed quizzes (3 quizzes), project-based homework assignments (3 assignments), and supplementary visual materials (e.g., a For Loop algorithm animation GIF) to make abstract concepts concrete.
- Conducted **one-on-one mentoring sessions** with participants struggling with hands-on tasks (inventory management and task tracker projects), guiding them through debugging and code structuring in a practical, real-world context.

Frontend Web Developer *2025*
Hacettepe University Kristal Geyik Awards Ankara, Turkey

- Redesigned and updated the official award ceremony website using **React.js**; applied modern, accessible, and mobile-responsive UI/UX principles to improve the event's public-facing presence.

Education

B.Sc. in Computer Engineering *2021 – Present*
Hacettepe University Ankara, Turkey

- Relevant coursework: Software Engineering, Distributed Systems, Machine Learning, Computer Vision, Database Management, Data Structures & Algorithms, Operating Systems.
- Active member and elected **Python instructor** at the IEEE Hacettepe Student Branch (2025–2026 academic term).